Upstox Bot: Complete Guide

What it is, how it works, the strategies, the research, and how it is deployed

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Version 1.1 · November 2026 · built by $um!t_K@Ktw@n · for Indian index options, stock options and shares (NSE / BSE) with free Upstox data
Project: github.com/Sumit211995/upstox_bot (private)

• Current mode: PAPER TRADING ONLY. No order code at all.

> **Current mode: PAPER TRADING ONLY.** This bot has **no order code at all**: it cannot place a trade on Upstox or anywhere else. It simulates every trade with real prices and the exact rules it would use with real money, and records the results. The optional Upstox token it uses is a read-only *Analytics* token that cannot place orders.

What changed in version 1.1 (13 Nov 2026)

Change	Details
Stocks tab: share swing trading	12-month momentum, top 3 shares, decided every 21 trading days, −15% stop, ₹20k paper account (chapter 6)
Live calls in market hours	09:15 Telegram call for the day's buys; instant alert when an open buy hits its stop or target (chapter 12)
Every call shows Entry · SL · Target · Time	On every card, table and Telegram message (chapter 7)
Option-chain recorder	Saves the live Nifty / Bank Nifty option chain every 5 minutes, building our own intraday history (chapter 10)
Live dashboard	Prices and the forming candle update every 2 s; closing-auction notice after 15:15 (chapter 11)
India VIX filter on stock-option buys	PF 1.51 → 1.99 on unseen 2025–26 data (chapter 8)
Tested and rejected	Buying at the close on a live call; a VIX filter for index buys and for selling; stock pullback and breakout rules (chapter 9)
Fixes	Server candle dates (pandas 3 stored dates in µs → 1970 dates); atomic cache writes shared by the two containers; share news searched by company name

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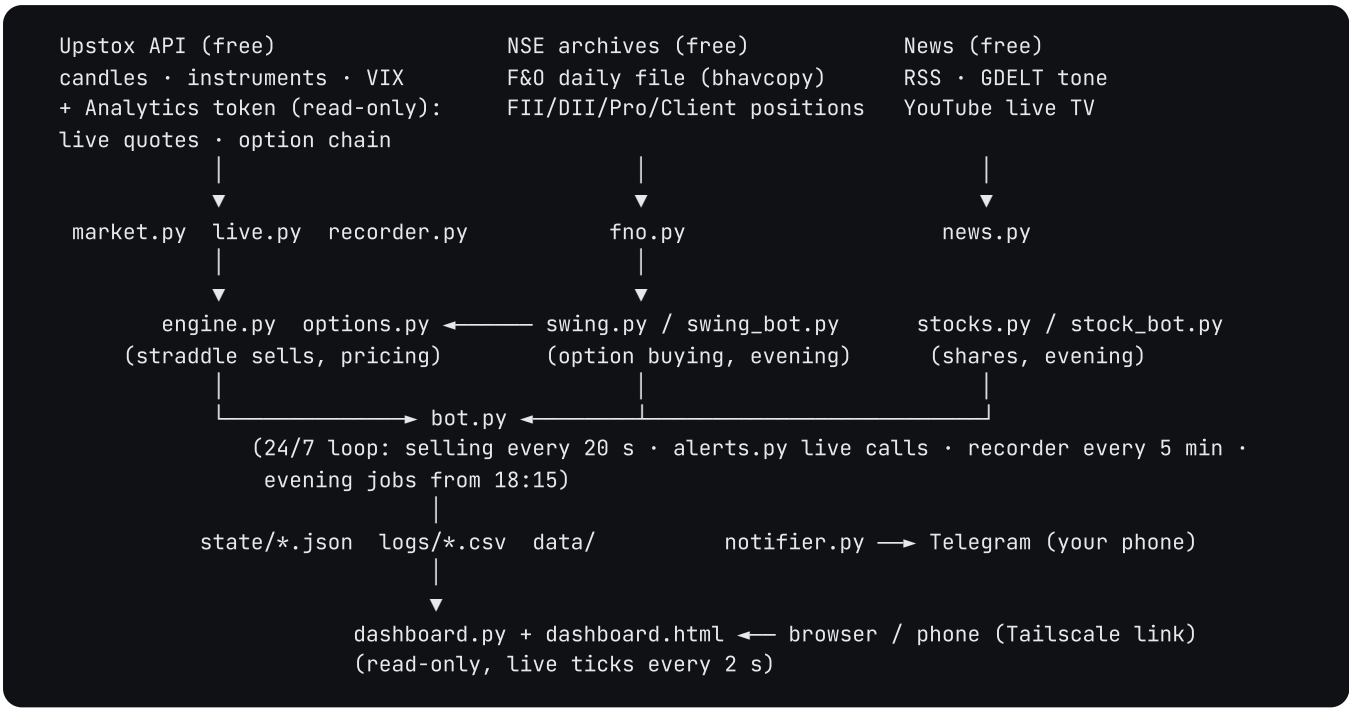
1. What the bot does, in one page

The bot paper-trades three things, because each needs a different amount of money and time.

	Selling (intraday)	**Option buying** (swing, 1–8 days)	**Shares** (swing, 1–3 months)
What	Sell the at-the-money call + put (a **straddle**), buy cheap **wings** 10 strikes away as insurance	Buy a call (CE) or put (PE), or a cheap **debit spread**, when a tested signal fires	Buy the **3 strongest large-cap shares** (12-month momentum)
Where	Nifty, Sensex, Bank Nifty, Midcap Nifty	Nifty, Bank Nifty and 20 large stocks (monthly options)	Nifty 50 + 20 large F&O stocks
When	Enter 10:00, stop per leg, exit 15:15	Signal after the close, buy at the next 09:15 open	Decided every 21 trading days at the close, bought at the next open
Money	Large margin (₹1 lakh+ per lot even hedged)	From **₹20,000**	From **₹20,000**
Why it earns	Premium decays to near zero by expiry; wings cap bad days	Follows big-money (FII/DII) positioning, breakouts, trend starts	Strong shares tend to keep outperforming for months
Paper records	6 books (`books.json`)	₹20k and ₹50k accounts (`buy_books.json`)	₹20k account (`stocks.py`)

The bot does **not** predict the future. Every rule was chosen on older data and then checked on newer data it never saw. Only rules that made money in **both** periods were kept. Everything stays **paper** until real results match the research (chapter 18).

2. The big picture: parts and how they connect



Two programs, one set of files. `bot.py` simulates every trade and writes `state/`, `logs/` and `data/`. `dashboard.py` only **reads** them plus public prices, so the page can never change anything. On the server each runs in its own Docker container; both share the same folders, so cache files are written atomically (a reader never sees a half-written file).

2.1 Tech stack at a glance

Layer	Technology	Where
Language	**Python 3.11** (bot, research, tests, web server) · HTML/CSS/JavaScript (dashboard page only)	all `*.py`, `dashboard.html`
Data and maths	**pandas**, **NumPy** (works on pandas 2 and 3)	everywhere
Option pricing	**Black-Scholes**, implied volatility, greeks (Δ, Γ, θ, vega), real Indian F&O charges	`options.py`
Market data	**Upstox public API v3** (index and share candles, instrument list); **Upstox Analytics token** for live quotes and the option chain	`market.py`, `live.py`, `stocks.py`
Exchange files	**NSE bhavcopy** (daily option prices, OI), **participant-wise OI** (FII/DII/Pro/Client)	`fno.py`
News	RSS (Economic Times, Moneycontrol, Business Standard, Mint, CNBC-TV18, NDTV Profit, BusinessLine), **GDELT** tone, YouTube live	`news.py`
Storage	**JSON** (state), **CSV** (journals, recorder), **pickle** caches; no database	`state/`, `logs/`, `data/`
Web dashboard	Python `http.server` + **TradingView Lightweight Charts**; read-only, strict security headers	`dashboard.py`, `dashboard.html`

Layer	Technology	Where
Alerts	**Telegram Bot API** (same bot as delta_bot)	`notifier.py`, `alerts.py`
Tests	**pytest**, 41 tests	`tests/`
Hosting	**Docker + Docker Compose** on **Oracle Cloud Always Free** (Mumbai), private access through **Tailscale**	`Dockerfile`, `docker-compose.yml`
Source	**Git + GitHub**, `Sumit211995/upstox_bot`; the server pulls from GitHub	

File	Lines	Job
`bot.py`	~310	Main loop: selling books every 20 s, live calls, recorder, evening jobs (options, then shares).
`engine.py`	~415	Selling simulator (hedged straddle, per-leg stop), expiry calendar, intraday research families.
`options.py`	~145	Black-Scholes, implied volatility, greeks, model volatility from VIX, Indian F&O charges.
`market.py`	~225	Upstox candles (5-minute since 2022, cached), intraday, VIX daily, instrument list, safe timestamps, atomic cache writes.
`live.py`	~155	Live quotes and option chain with the token; live estimate for open buys without it.
`fno.py`	~245	NSE F&O daily files and participant positions, cached.
`swing.py`	~655	Option-buying features, signal families, trade simulator, combined accounts.
`swing_bot.py`	~355	Option-buying evening job: tonight's signals and contracts, paper accounts, Telegram plan.
`stocks.py`	~555	Share data (daily since 2018), indicators, delivery charges, simulator, momentum ranking, paper account.
`stock_bot.py`	~150	Share evening job: paper step, ranking, news, Telegram calls.
`alerts.py`	~130	09:15 calls and the live stop / target watch.
`recorder.py`	~85	Option-chain snapshots every 5 minutes.
`news.py`	~210	Headlines per channel / symbol / company, mood score, GDELT tone, live TV ids.
`dashboard.py` + `dashboard.html`	~500 + ~1,350	The web page and its read-only API.
`books.json`, `buy_books.json`		Selling books; option-buying strategies and accounts.
`research/`		Every research script and its results (`research/out/`).

3. One day of the bot, hour by hour

Time (IST)	What happens
09:15	Market opens. Prices update every **2 s**. **Live call** on Telegram: today's option buys and share orders (bought at this open).
09:15–15:15	Every 5 min: option-chain snapshot saved. Every minute: open option buys are priced live; a **stop or target hit is sent at once**.
10:00	Selling books enter: sell ATM CE + PE, buy wings 10 strikes away (only on the days each book trades).
10:00–15:15	Every 20 s: each short leg checked against its stop; a stopped leg is closed, the other stays.
15:15	Selling books close everything. Indices freeze for the **closing auction** (since Aug 2026) until ~15:30.
15:30	Market closes. Day summary to Telegram.
18:15 →	Evening jobs (retry every 15 min until NSE publishes, last try 23:45): **shares** (paper step, ranking, calls) and **option buying** (stops / targets / time exits on real NSE prices, tonight's signals and contracts). New calls go to Telegram.

4. Selling strategy: hedged short straddle

The trade. At 10:00 sell the **at-the-money** call and put of the nearest expiry and buy the call and put **10 strikes further out** (the wings). You collect premium; the wings cap the loss if the index runs far in one direction.

The exits. Each sold leg has its own stop: if its premium rises by the book's stop % (60–100%), that leg is bought back. At 15:15 everything is closed. **Intraday only.**

Book	Days traded	Stop per sold leg
Nifty · expiry day	Weekly expiry (Tuesday)	80%
Nifty · every day	Every day of the weekly cycle	100%
Sensex · expiry day	Weekly expiry (Thursday, BSE)	60%
Sensex · every day	Every day of the weekly cycle	80%
Bank Nifty · expiry day	Monthly expiry only (≈ 1 trade a month)	100%
Midcap Nifty · expiry day	Monthly expiry only	100%

5. Option buying: four signals for a small account

All option-buying signals are computed **after the close** from the day's final data, and bought at the **next morning's open**. Buying at the close on a live call instead was tested and lost money (chapter 9).

#	Signal	Rule	Contract	SL · Target · Time
1	**Index smart money**	Average 60-day rank of five NSE participant measures (DII futures share, DII futures change, FII options net, *against* DII options net, *against* retail futures change). Top 20% → CE, bottom 20% → PE.	Nifty / Bank Nifty ATM, ≥ 25 days to expiry	−50% · none (exit day 5) · swing ≤ 5 days
2	**Index 20-day breakout (1:2)**	Close above the 20-day high → CE; below the 20-day low → PE.	1 strike in the money	−35% · +70% · swing ≤ 8 days
3	**Index EMA 9/21 cross (1:2)**	EMA 9 crosses EMA 21 in the direction of EMA 50.	1 strike in the money	−25% · +50% · swing ≤ 3 days
4	**Stock smart money**	Same market-wide score on 20 large F&O stocks; **skipped when India VIX is in its top 20% of 60 days**.	Monthly stock option, 1 strike ITM	−30% · none (exit day 5) · swing ≤ 5 days

Shown but **not traded**: the **OI-extreme** signal (the "30:70 open interest" idea as a 60-day extreme of call:put OI within ±2% of the price). It works only with one exact setting, so it is not trusted.

Why smart money works. NSE publishes every evening how many futures and options contracts **FII** (foreign funds), **DII** (Indian funds), **Pro** (brokers' own) and **Client** (retail) hold. Following DII futures and FII options, and going *against* retail futures, pointed the right way. One group alone did not (FII futures alone: PF 0.70).

Greeks. Cards show **θ** (₹ lost per day from time passing) and **Δ** (move per 1 index point) as information. The ₹20k account uses **debit spreads** because they cut θ (e.g. −₹790/day plain vs −₹138/day as a 6-strike spread).

6. Stocks: share swing trading (momentum)

The rule (`stocks.FINAL`).

1. Universe: the Nifty 50 + the 20 large F&O stocks, daily prices since 2018 (already split / bonus adjusted).
2. Every **21 trading days** at the close, rank the shares by their return from **12 months ago to 1 month ago**, keeping only shares above their **200-day EMA**.
3. Hold the **top 3**, equal weight, whole shares only. Sell anything that dropped out of the top 3.
4. **Market filter:** if Nifty is below its own 200-day EMA on a decision day, sell everything and stay in cash.
5. **Stop loss −15%** below the entry. **No fixed target**: a winner is held while it stays in the top 3.
6. Orders decided at the close are filled at the **next 09:15 open**. Delivery charges (STT, stamp, exchange, GST, ₹20 DP charge per sale) and 0.1% slippage are included.

Why it works. Shares that have risen strongly for a year tend to keep outperforming for several more months (momentum). Skipping the last month avoids buying a short-term spike; the market filter avoids holding shares through a broad bear market.

Read the limits. Most of the gain came in 2021 and 2024; it trailed Nifty in 4 of 8 years; the share list is **today's** Nifty 50, which includes past winners and flatters history; only 26 trades in the unseen period.

7. Every call: Entry · SL · Target · Time

Every buy or sell call shows the same four fields: on every dashboard card, in the signals tables and in every Telegram message.

Call	Entry	SL	Target	Time
Index option buy	next 09:15 open (≈ last close price)	option price −50 / −35 / −25%	+70 / +50%, or none (exit on day N)	Swing · ≤ 5 / 8 / 3 days
Stock option buy	next 09:15 open	−30%	none (exit on day 5)	Swing · ≤ 5 days
Straddle sell	10:00 · credit ₹	each sold leg's stop price	time decay · exit 15:15	**Intraday**
Share buy	next 09:15 open (ref = last close)	−15% from entry	none fixed (top-3 rule)	**Swing · 1–3 months**
Open holdings	your entry price and date	live stop	target	days held so far

Example Telegram call:

```
📈 STOCK CALL · tomorrow's open (paper, ₹20,000 momentum account)
• BUY SHRIRAMFIN
  Entry: tomorrow's open (ref ₹1,003.80)
  SL: ₹853.23 (-15% from entry)
  Target: none fixed: held while in the top 3 (reviewed every 21 trading days)
  Time: Swing · about 1-3 months
```

8. Research results: what the data says

Method. Real NSE daily option prices for option buying; modelled premiums checked against NSE prices for intraday selling; real daily share prices for shares. Lot sizes, brokerage, STT, exchange + SEBI charges, stamp duty, GST and slippage are included. **Settings chosen on the older period, judged on the newer one.** If a stop and a target were both touched on the same day, the **stop is assumed first**.

8.1 Selling books (1 lot; tuned 2022–24, unseen 2025–26)

Book	2022–24: trades · win · PF	2025–26: trades · win · PF · net
Nifty · expiry day	151 · 60% · 2.02	**101 · 58% · 2.50 · +₹1.25 L**
Nifty · every day	723 · 69% · 1.91	**461 · 66% · 1.96 · +₹2.56 L**
Sensex · expiry day	80 · 70% · 2.37	**91 · 76% · 1.66 · +₹65k**
Sensex · every day	405 · 60% · 1.79	**461 · 56% · 1.44 · +₹95k**
Bank Nifty · expiry day	155 · 52% · 2.47	**29 · 59% · 3.48 · +₹62k**
Midcap Nifty · expiry day	149 · 68% · 4.75	**24 · 71% · 2.23 · +₹39k**

8.2 Option-buying signals (each alone; tuned 2022–24, unseen 2025–26)

Signal	2022–24: trades · win · PF	2025–26: trades · win · PF
Index smart money	18 · 67% · 3.85	**11 · 55% · 4.06**
Index 20-day breakout (1:2)	36 · 53% · 2.27	**20 · 50% · 2.03**
Index EMA cross (1:2)	15 · 60% · 3.01	**15 · 40% · 1.31**
Stock smart money (with VIX filter)	192 · 51% · 2.36	**117 · 50% · 1.95**
OI-extreme (shown only)	149 · 37% · 1.92	73 · 32% · 1.67

India VIX filter (stock options only): skipping stock-option buys when VIX is in its top 20% of 60 days raised PF 1.87 → 2.36 (2022–24) and 1.51 → 1.99 (2025–26), keeping ~70% of trades.

8.3 Combined option accounts

Account	2025–26 (unseen)	Start in any month, run 12 months (47 runs)
₹20k (spreads)	37 trades · 60% win · PF 2.1 · ₹20k → ₹58k · lowest ₹15.3k	profit in **74%** · median end ₹27.4k · fell below half 9%
₹50k	46 trades · 48% win · PF 2.4 · ₹50k → ₹2.38 L	profit in **87%** · median end ₹1.18 L · fell below half 6%

8.4 Shares (₹20k account, max 3 positions; tuned 2019–23, unseen 2024–Nov 2026)

Family	2019–23: trades · win · PF · CAGR · worst drop	**2024–26: trades · win · PF · CAGR · worst drop**
Trend pullback (RSI 2 dip above EMA 200)	295 · 26% · 0.26 · −50% · −97%	206 · 24% · 0.14 · −44% · −81%
Breakout (20-day high, volume, ATR stop, 1:2)	191 · 44% · 1.06 · +3% · −48%	110 · 36% · 0.63 · −15% · −43%
Momentum, top 3, Nifty > EMA 200, −15% stop	**64 · 50% · 1.85 · +18.9% · −28%**	**26 · 50% · 4.55 · +23.0% · −18%**
Nifty 50 buy and hold	+14.8% a year · −38%	+5.4% a year · −16%

Stop loss test (momentum): none → PF 1.70 / 4.55; **−15% → 1.85 / 4.55 (kept: better on 2019–23, same on 2024–26)**; −10% → 1.99 / 3.78 (rejected: worse on 2024–26); −20% / −25% → no gain on 2024–26.

Year	2019	2020	2021	2022	2023	2024	2025	2026*
Momentum (₹20k)	−4.5%	+4.1%	+73.5%	+19.9%	+7.9%	+96.4%	−2.9%	+2.3%
Nifty 50	+12.0%	+14.9%	+24.1%	+4.3%	+20.0%	+8.8%	+10.5%	−3.2%

*2026 to mid-November.

> **Read these numbers honestly.** Index smart money has only **11** unseen trades and the share strategy **26**: real results could be much lower. One in four 12-month runs of the ₹20k option account still lost money. Paper first.

9. What did not work (tested and rejected)

Idea	Result
Intraday option **buying** (opening-range breakout, Supertrend, pullback) on any index	Lost in both periods on all 4 indices (unseen PF 0.5–0.8)
Buying at the close on a live call (breakout / EMA signals seen from the live price at ~15:10)	Lost: breakout PF 2.03 → 0.91, EMA 1.31 → 0.50 (2025–26). Entry stays at the next open.
Plain stock option buying on price signals	Lost in ~99.6% of settings
A fixed **30:70 OI ratio**	Almost never happens on indices; the relative version is only shown
FII futures long % alone	PF 0.70
News as a filter	Made results worse (PF 1.44 → 1.13); shown as context only
RSI, MACD, Supertrend, Bollinger squeeze as option signals	Failed on unseen data
VIX filter on index option buys	No setting improved both periods
Day filters for selling (ADX, RSI, smart money, squeeze, VIX)	None improved both periods without cutting total profit ("skip VIX < 12" raised PF but lowered profit)
Share **pullback** and **breakout** rules	Lost after delivery charges in both periods
Share momentum + VIX filter / + smart-money filter / without the market filter	Worse or no better on unseen data
Full-lot option buying with ₹20k	Only 47% of 12-month runs profitable → debit spreads instead

10. Data sources (all free) and the option-chain recorder

Data	Source	Used for
Index candles (5-minute, since Jan 2022)	Upstox public historical API	Selling simulator, charts
Share candles (daily, since 2018)	Upstox public historical API	Share research, Stocks tab, charts
Live quotes, option chain + greeks	Upstox **Analytics token** (read-only, 1 year)	Live ticks, live marks, stop / target alerts, recorder
Daily option prices + open interest	NSE F&O bhavcopy	Option-buying research and evening job
FII / DII / Pro / Client positions	NSE participant-wise OI	Smart-money score
FII / DII cash buying	NSE (latest day)	Big money panel
India VIX	Upstox	Stock-option filter, option model
Company names	Upstox instrument file	Share news ("Titan Company", not "Titan Mining")
Headlines	RSS feeds of 7 business outlets + Google News	News panel, share news
News tone	GDELT	Context only
Live business TV	YouTube (CNBC-TV18, ET Now, NDTV Profit, Zee Business, CNBC Awaaz)	News & TV tab

Option-chain recorder (`recorder.py`). NSE's free files are daily only, so intraday option strategies could never be tested. Every 5 minutes in market hours the bot now saves the live Nifty and Bank Nifty option chain (2 nearest expiries, strikes within ±8%: price, volume, open interest, bid/ask, IV and greeks) to `data/chain/<INDEX>/<day>.csv.gz` (≈ 2 MB a day). After 2–3 months this becomes our own intraday history for open-interest research.

How to get the Upstox Analytics token: Upstox developer portal → *Apps* → **Analytics token** → generate (valid one year) → put it in `.env` as `UPSTOX_ANALYTICS_TOKEN=` on the server and restart. Free and read-only.

11. The dashboard

Tabs: **Overview · Buying · Selling · Stocks · Big money · News & TV.**

- **Price tiles** update every 2 s and flash on each change. The pill shows *Live · Upstox* (real quotes) or *Live · ~1 min* (no token). From 15:15 to 15:30 it shows the closing-auction pause.
- **Chart** (5m / 15m / 1h / 1D): the last candle moves with the live price. Buying tab: buy arrows of the top 3 buying strategies; Selling tab: straddle trades of the top selling books.
- **Signal cards** (Overview, Buying, Selling): every call with its Entry · SL · Target · Time block and the strategy's success rate, best first. Share calls appear in the Overview strip too.
- **Buying tab:** ₹20k / ₹50k accounts, tonight's signals, ranked strategies, all trades, OI scanner.
- **Stocks tab:** account, calls for the next open, holdings with live prices (SL, target, days held), today's momentum top 10 (tap a share for its 2-year chart with EMA 50 / 200, buy / sell arrows and news), the research by year vs Nifty, closed trades.
- **Big money:** FII / DII / Pro / Client positions and ranks; FII / DII cash.
- **News & TV:** headlines per channel and live TV inside the page.

12. Live calls and Telegram alerts

Message	When
🇮🇳 PAPER ENTRY / EXIT (straddle sells)	Live: 10:00 entry, a leg's stop at any moment, 15:15 exit
📋 PLAN (option buys)	Evening, for the next 09:15 open
📈 STOCK CALL (share buys / sells)	Evening, for the next 09:15 open
🔔 LIVE CALL · today's buys	09:15, repeats the evening's option and share calls
🛑 STOP hit → exit / 🎯 TARGET hit → book profit	Live, the minute an open option buy reaches its level (real Upstox price)
Fills, exits, day summary	As they happen / after the close

Each alert is sent once (remembered across restarts). Paper accounts are settled each evening on NSE's official prices; the live alerts are the "act now" calls for anyone trading the same idea for real.

13. Money: the paper accounts, with worked examples

Account	Capital	Per trade	Max at once
Option buying ₹20k (recommended start)	₹20,000	≈ ₹7,000 (index signals as debit spreads)	2
Option buying ₹50k	₹50,000	≈ ₹15–20k (plain CE / PE)	2
Shares ₹20k	₹20,000	≈ ₹6,600 (equity ÷ 3, whole shares)	3

Option example (₹20k, illustrative). Smart-money score hits its top 20% → buy Nifty 25,300 CE and sell 25,600 CE. Spread ≈ ₹110 × 65 = ₹7,150. SL −50% (≈ ₹3,600 loss + charges). Exit by day 5 at the latest. Maximum value = width 300 × 65 = ₹19,500.

Share example (₹20k). Call: BUY Shriram Finance, ref ₹1,003.80. At the next open the account spends ≈ ₹6,600: 6 shares at ≈ ₹1,004 (+ charges). SL ₹853 (−15%). Held while it stays in the top 3; reviewed every 21 trading days.

14. Security

Rule	How
No trading possible	There is no order code; the token is read-only
Secrets stay private	`.env` is never committed (`.gitignore`), `chmod 600` on the server; the token is only sent to `api.upstox.com`
Public repo, private server	The GitHub repo contains no IPs, tokens or keys
Page cannot be attacked from outside	Bound to `127.0.0.1` ; reachable only through your private Tailscale network
Page hardening	Content-Security-Policy, `X-Frame-Options: DENY` , `nosniff` , only GET allowed, parameter allowlist, http(s)-only links

15. Deployment: Oracle Cloud + Tailscale

The bot runs next to delta_bot on the same **Oracle Cloud Always Free** VM (Mumbai, 1 GB RAM + 2 GB swap).

	delta_bot	upstox_bot
Folder	`~/delta_bot`	`~/upstox_bot` (a clone of GitHub)
Containers	`delta-bot`, `dashboard`	`upstox-bot`, `upstox-dashboard`
Dashboard (on the server)	`127.0.0.1:8000`	`127.0.0.1:8002`
Link (phone / laptop with Tailscale)	`https://<your-server>.<tailnet>.ts.net`	`https://<your-server>.<tailnet>.ts.net:8443`

Open it on your phone: install the Tailscale app, sign in with the same account, open the link.

Update after a code change:

```
# on the laptop
git push
# on the server
ssh -i ~/Downloads/oracle.key ubuntu@<server-ip>
cd ~/upstox_bot && git pull && docker compose up -d --build
```

`.env`, `state/`, `logs/` and `data/` are not in Git and are never touched by an update.

Memory and disk. The two containers use ≈ 230 MB. The option evening job briefly needs more and runs ~10–15 minutes on the small server (swap); the share job takes ~20 s. Never start a second evening job by hand while the bot's own one is running. The recorder adds ≈ 2 MB a day (~0.5 GB a year).

Where to check it is running: the Oracle console only shows the VM (CPU / memory). The bots: the dashboard link, Telegram "bot started", or on the server `docker compose ps` and `docker compose logs --tail 30`.

16. Daily use: commands

Command	What it does
`python bot.py`	Run the bot (loop)
`python bot.py --status`	Paper results so far
`python bot.py --evening`	Run the option-buying evening job now (only if the bot is not already running it)
`python -c "import stock_bot; stock_bot.run()"`	Run the share job now (also only when the bot is not running it)
`python dashboard.py`	Dashboard on http://localhost:8002
`python -m pytest -q`	Run the 41 tests
`python research/buy_final.py`	Rebuild the option-buying history shown on the dashboard
`python research/stock_final.py`	Rebuild the share research shown on the Stocks tab
`python docs/build_pdf.py`	Rebuild this PDF

17. Configuration reference (`.env`)

Key	Default	Meaning
`INDICES`	`NIFTY,BANKNIFTY,SENSEX,MIDCPNIFTY`	Indices shown and traded by the selling books
`PAPER_CAPITAL`	`200000`	Paper capital per selling book (₹)
`LOTS`	`1`	Lots per selling trade
`POLL_SECONDS`	`20`	Check interval in market hours
`TELEGRAM_TOKEN` , `TELEGRAM_CHAT_ID`	empty	Telegram alerts (optional)
`UPSTOX_ANALYTICS_TOKEN`	empty	Live quotes, option chain, live alerts, recorder (optional, read-only)

Put comments on **their own line** in `.env` , never after a value (`KEY= # note`): Docker reads the comment as the value.

18. The plan: from paper to real money

1. **Paper for 1–2 months** (now). Watch the option accounts, the selling books and the share account.
2. **Compare with the research.** Go further only if paper results look roughly like chapter 8: at least ~15 option buys and a few share rebalances, profit factor above ~1.3, no rule surprises.
3. **Start small.** Real option buying with ₹20k and/or the share account with ₹20k, only for the parts that held up on paper; selling only with enough margin.
4. **Intraday open-interest research** once the recorder has 2–3 months of option-chain history.
5. **Real orders are not built.** They would need Upstox's order API with a separate trading token, plus a safety layer like delta_bot's (daily loss limit, kill switch). That is a deliberate later step.

19. Troubleshooting and FAQ

Question	Answer
Index price stops moving at 15:15	Normal since Aug 2026: closing auction. VIX keeps moving.
No option buy signals tonight	Normal: they fire a few times a month.
Stocks tab says "Next decision: tonight" / no calls	The share account decides every 21 trading days (or when the market filter flips); calls appear after that evening's run.
Why does a share call have no target?	The tested rule keeps a winner while it stays in the top 3; a fixed target cut the big winners. It has a −15% stop.
Why no option buys on Sensex / Midcap?	Option-buying research used NSE data (Nifty, Bank Nifty); Sensex is on BSE.
Dashboard says "Live · ~1 min"	The Upstox token is missing or expired (one year); generate a new one.
An evening job did not run	NSE publishes late some days; the bot retries every 15 minutes until 23:45.
Server CPU near 100% in the evening	The option evening job; it finishes in ~15 minutes.
Paper trades are not in my Upstox app	Correct: they are simulated; see the dashboard and Telegram.
Old dashboard on the laptop shows no trades	The bot runs on the server now; use the Tailscale link.

20. Glossary

Term	Meaning
CE / PE	Call option (profits when the price rises) / put option (profits when it falls)
ATM / ITM / OTM	Strike at / inside / outside the current price
Straddle	Selling (or buying) a CE and a PE at the same strike
Wings	Far-away options bought as insurance on a short straddle
Debit spread	Buy one option, sell a cheaper one further out: lower cost and decay, capped profit
Momentum	Ranking shares by how much they rose over the past 12 months (skipping the last month)
EMA 200	200-day exponential moving average: above it = long-term uptrend
Rebalance / decision day	Every 21 trading days the share account re-ranks and swaps holdings
Delivery	Buying shares to hold for days or months (not intraday)
Expiry	Last day of an option contract
OI	Open interest: contracts still open
FII / DII / Pro / Client	Foreign funds / Indian funds / brokers' own trading / retail
PF (profit factor)	Total won ÷ total lost. Above 1 = profitable
CAGR	Average yearly growth of the account
Walk-forward	Choose settings on old data, judge on newer unseen data
VIX	India's volatility index: high = options expensive
θ theta / Δ delta	Daily time decay / move per 1 point of the underlying
Bhavcopy	NSE's daily file of every contract's prices and open interest
Closing auction	Since Aug 2026, NSE/BSE fix closing prices in an auction after 15:15; indices pause until ~15:30